

MIDRAND GRAND CENTRAL, South Africa

WMO: 682650

Lat: 25.986S

Lon: 28.140E

Elev: 1621

StdP: 83.31

Time Zone: 2.00 (E02)

Period: 05-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.2	2.9	-14.2	1.3	13.5	-12.8	1.5	13.4	8.6	6.3	7.9	8.6	5.1	170	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.7	29.1	14.4	28.1	14.3	27.1	14.5	18.2	23.6	17.8	23.2	17.2	22.5	3.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	14.5	19.6	16.1	14.0	19.4	15.8	13.7	19.3	58.3	23.5	56.8	23.3	55.4	22.9	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.5	7.5	6.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-3.2	19.4	1.0	0.7	-4.0	19.8	-4.6	20.2	-5.2	20.6	-5.9	21.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	19.9	20.1	18.3	15.6	14.8	11.5	11.2	12.8	17.2	18.6	18.7	19.3
	DBStd	4.14	2.17	2.06	2.69	2.74	2.84	2.64	2.50	3.53	3.28	2.88	2.90	2.24
	HDD10.0	50	0	0	0	1	4	15	17	13	1	0	0	0
	HDD18.3	980	11	7	32	86	113	204	221	173	58	32	29	16
	CDD10.0	2422	306	284	258	170	152	62	54	100	217	267	262	289
	CDD18.3	310	58	58	32	5	3	0	0	2	24	41	41	47
	CDH23.3	2124	298	331	167	29	15	0	0	18	235	413	329	289
	CDH26.7	321	44	51	17	0	0	0	0	24	88	62	36	
Wind	WSAvg	3.3	3.3	3.2	3.0	2.8	2.7	3.1	3.0	3.5	3.7	4.0	3.9	3.6
Precipitation	PrecAvg	691	125	104	96	39	16	7	2	4	15	60	96	127
	PrecMax	1123	231	309	327	117	98	33	13	30	72	167	199	190
	PrecMin	448	45	23	17	1	0	0	0	0	0	10	18	44
	PrecStd	168	51	67	68	29	23	9	3	7	19	41	42	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.8	30.1	28.8	25.8	25.0	21.2	21.1	25.2	28.7	30.2	30.1	29.1
		MCWB	15.3	15.7	15.3	12.9	11.3	9.2	8.0	10.4	11.2	13.5	14.9	15.5
	2%	DB	27.9	28.2	26.8	24.1	23.2	19.9	19.8	23.1	27.2	29.0	28.2	27.8
		MCWB	16.0	15.4	15.0	12.7	10.3	8.3	7.8	10.0	11.1	12.8	14.5	15.4
	5%	DB	26.2	26.9	25.2	22.9	22.0	18.8	18.2	21.2	26.0	27.2	26.9	26.2
		MCWB	16.5	15.8	14.9	12.8	10.3	8.0	7.4	9.2	11.2	13.1	14.8	15.6
	10%	DB	25.0	25.2	23.9	21.2	20.8	17.1	17.0	19.9	24.8	25.8	25.1	25.0
		MCWB	16.8	15.9	15.1	12.4	10.4	7.6	7.3	8.5	10.8	12.9	14.1	15.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	18.6	18.0	16.6	14.5	10.9	10.2	12.0	13.9	16.4	17.4	18.5
		MCDB	24.1	23.6	22.4	21.4	19.7	17.7	16.5	21.5	22.8	23.7	24.8	24.4
	2%	WB	18.2	18.0	17.2	15.5	13.0	10.0	9.3	11.2	13.1	15.6	16.7	17.8
		MCDB	23.5	23.7	21.9	19.9	18.2	16.5	16.1	20.1	21.7	23.2	23.6	23.3
	5%	WB	17.9	17.3	16.7	14.5	12.1	9.4	8.5	10.3	12.4	15.0	16.1	17.2
		MCDB	23.2	23.1	21.5	19.1	18.2	15.8	15.8	18.3	21.6	22.7	23.1	22.5
	10%	WB	17.4	16.9	16.3	13.7	11.4	8.7	7.8	9.6	11.7	14.4	15.5	16.7
		MCDB	22.3	22.6	21.2	18.7	17.8	15.0	15.4	17.3	21.5	21.8	22.1	21.9
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.7	10.1	10.3	10.8	10.9	11.2	11.9	13.0	12.7	11.1	10.5
		MCDBR	11.5	12.4	11.8	11.6	11.5	11.5	11.9	12.4	13.5	14.0	12.8	12.1
		MCWBR	4.1	5.0	4.1	4.3	4.8	5.3	5.3	5.2	5.4	4.9	4.6	4.5
	5% WB	MCDBR	9.4	10.4	9.4	9.6	10.0	9.9	11.0	12.0	13.0	12.4	11.2	10.0
		MCWBR	4.0	4.4	4.0	4.4	4.9	4.9	5.3	5.7	5.6	5.0	4.7	4.4
Clear-Sky Solar Irradiance	taub		0.384	0.386	0.375	0.328	0.293	0.275	0.276	0.329	0.369	0.402	0.364	0.371
	taud		2.333	2.326	2.337	2.440	2.518	2.542	2.525	2.368	2.270	2.213	2.359	2.359
	Ebn at Noon		959	941	922	922	918	916	927	907	912	916	972	973
	Edh at Noon		136	134	126	103	87	80	85	109	132	148	132	133
All-Sky Solar Radiation	RadAvg		6.63	6.28	5.56	4.77	4.36	3.96	4.30	5.07	6.03	6.47	6.74	6.79
	RadStd		0.69	0.56	0.51	0.42	0.22	0.20	0.11	0.23	0.28	0.44	0.61	0.40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.66	+0.75	N/A	N/A	N/A	N/A	N/A	-181	+234	+87

Nomenclature: See separate page